Activity 3

ICEPOT Framework – Prompt Makeover Challenge

Objective

Understand how a weak or ambiguous QA prompt can be transformed using the ICEPOT framework to produce a more specific, structured and business-aligned AI response.

ICEPOT

I – Instruction — What should the AI do?

C – Context — What background information does it need?

E – Example — What reference should it follow?

P – Persona — What role should it assume?

O – Output Format — How should the response be presented?

T – Tone — How should it be communicated?

Scenario 01 – Search Functionality

1. Weak Prompt

"Test the search feature."

2. What Information Is Missing?

• The weak prompt is missing the complete ICEPOT structure.

• It does not define the application/module, persona, reference example, output format, audience or tone.

3. ICEPOT Prompt

Act as a **Senior Quality Assurance Engineer** and test the search feature.

This scenario is applicable to the **Lead List module in the TAN CRM application**.

Use the provided Admin Login Gherkin feature as a reference for the expected Gherkin structure, tagging and style.

Create a **Gherkin feature file** for the Lead List search functionality. It must be understandable to Business Analysts, Developers and Quality Analysts, including technical and non-technical team members. Keep scenarios at business-behavior level and avoid locators, API calls, database queries and automation-specific actions. Include positive and negative scenarios using Given/When/Then. Keep smoke coverage focused on the critical journey. Use clear, concise and professional language.

4. ICEPOT Response – Summary

The response generated a specific Gherkin feature for **Search leads in Lead List**, with smoke/lead-list tags, login and Lead List background, valid search, multiple matches, no matches, empty search and clear-search scenarios. It also explained why business-level Gherkin is preferable to implementation-specific steps.

5. Three Comparison Observations

1. The weak prompt is missing the whole ICEPOT structure.
2. The ICEPOT prompt response is more specific than the weak prompt response and is closer to what the user/business expects.
3. Defining the audience and output format resulted in business-readable Gherkin scenarios rather than a generic QA checklist.

Scenario 02 – File Upload

1. Weak Prompt

"Generate test cases for file upload."

2. What Information Is Missing?

• The prompt does not define what aspects of file upload should be tested.

• It lacks application context, supported file types, size limits, validation rules, persona, output format, audience and tone.

3. ICEPOT Prompt

Act as a **Senior Quality Assurance Engineer** and generate test scenarios for a **file upload functionality** in a business application.

Cover valid uploads, unsupported file types, file size limits, empty files, duplicate files, invalid/corrupted files, upload cancellation, upload failure and successful completion.

Organize the scenarios for Business Analysts, Developers and Quality Analysts. Provide a structured table with **Test Case ID, Scenario, Test Data, Steps, Expected Result and Priority**. Use clear, concise and professional QA language.

4. ICEPOT Response – Summary

The response provides a structured file-upload test-case set covering valid and invalid files, validation rules, boundary conditions, duplicates/corruption and upload failures using the requested format.

5. Three Comparison Observations

1. The weak prompt leaves the testing scope and business context ambiguous.
2. The ICEPOT prompt makes expected file-upload coverage more specific, including positive, negative and boundary scenarios.
3. Defining the output structure makes the result easier for QA, Developers and Business Analysts to review.

Scenario 03 – Shopping Cart

1. Weak Prompt

"Test the cart functionality."

2. What Information Is Missing?

• The prompt does not define the application context, cart operations or business rules.

• It lacks positive/negative scope, persona, output format, audience and tone.

3. ICEPOT Prompt

Act as a **Senior Quality Assurance Engineer** and test the **Shopping Cart functionality** of an e-commerce application.

The cart allows users to add products, remove products, update quantities and proceed toward checkout.

Consider adding/removing products, quantity updates, price and total calculation, stock availability, cart persistence and empty-cart behavior. Include positive and negative scenarios. Provide **Test Case ID, Scenario, Preconditions, Steps, Test Data, Expected Result and Priority**. Make the scenarios understandable to Business Analysts, Developers and Quality Analysts.

4. ICEPOT Response – Summary

The response provides structured shopping-cart test cases covering add/remove, quantity updates, pricing, stock availability, persistence and empty-cart behavior with positive and negative coverage.

5. Three Comparison Observations

1. The weak prompt does not provide the ICEPOT context needed to define expected cart behavior.

2. The ICEPOT prompt produces a more specific scope covering add, remove, quantity, pricing, stock and persistence behavior.

3. Defining positive/negative coverage and output format makes the testing approach clearer to the whole team.

Scenario 04 – Role-Based Access Control

1. Weak Prompt

"Write test cases for user roles."

2. What Information Is Missing?

• The prompt does not define application context, roles, permissions or authorized/unauthorized behavior.

• It lacks security expectations, persona, output format, audience and tone.

3. ICEPOT Prompt

Act as a **Senior Quality Assurance Engineer** and create test cases for **Role-Based Access Control (RBAC)** in a business application.

The application has multiple user roles, and each role should access only permitted features and actions.

Validate authorized and unauthorized access, page access, feature visibility, create/update/delete permissions, direct URL access and behavior after role changes. Include positive and negative scenarios. Provide **Test Case ID, Role, Scenario, Preconditions, Steps, Expected Result and Priority**.

4. ICEPOT Response – Summary

The response provides role- and permission-oriented test cases covering authorized and unauthorized actions, direct access attempts, feature visibility and role-change behavior.

5. Three Comparison Observations

1. The weak prompt does not define roles, permissions or access behavior.
2. The ICEPOT prompt makes the scope more specific by including authorized, unauthorized and security-related scenarios.
3. Defining persona and output structure makes the test cases more useful for functional and security validation.

Scenario 05 – API Testing

1. Weak Prompt

"Write API test cases for user registration."

2. What Information Is Missing?

• The prompt does not specify API context, request method, required fields, validation rules or expected responses.

• It lacks duplicate-registration behavior, error handling, status-code expectations, persona and output format.

3. ICEPOT Prompt

Act as a **Senior QA Engineer specializing in API testing** and create test cases for a **User Registration API**.

The API accepts registration details such as name, email and password. Cover positive, negative, boundary and validation scenarios including valid registration, missing mandatory fields, invalid formats, duplicate email, password rules, unsupported values, malformed requests and error handling. Validate HTTP status codes, response body and messages. Provide **Test Case ID, Scenario, HTTP Method, Request/Test Data, Expected Status Code, Expected Response and Priority**.

4. ICEPOT Response – Summary

The response provides API-specific test cases covering valid/invalid registration, missing fields, duplicate users, boundary/validation cases, malformed requests, status codes and response bodies.

5. Three Comparison Observations

1. The weak prompt does not provide API-specific context and validation requirements.
2. The ICEPOT prompt provides more specific positive, negative, boundary and validation scenarios.
3. Defining HTTP method, request data, expected status code and response makes the output more useful for API testing.

Scenario 06 – Mobile App – OTP Login

1. Weak Prompt

"Test OTP login on mobile."

2. What Information Is Missing?

• The prompt does not define the OTP flow, expiry rules, retry limits or mobile-specific behavior.

• It lacks network conditions, resend behavior, persona, output format, audience and tone.

3. ICEPOT Prompt

Act as a **Senior Mobile QA Engineer** and test **OTP-based login** in a mobile application.

The user enters a registered mobile number, receives an OTP and is authenticated when the valid OTP is entered.

Cover valid, invalid and expired OTPs, incorrect attempts, resend OTP, multiple requests, maximum retry attempts, unregistered numbers, network interruption and successful login. Consider app backgrounding/foregrounding and temporary network loss. Provide **Test Case ID, Scenario, Preconditions, Steps, Test Data, Expected Result and Priority**.

4. ICEPOT Response – Summary

The response provides mobile-specific OTP test cases covering valid/invalid/expired OTPs, resend, retry limits, unregistered numbers, network interruption and app state changes.

5. Three Comparison Observations

1. The weak prompt does not provide sufficient OTP-flow or mobile-specific information.
2. The ICEPOT prompt provides a more specific set of positive, negative and mobile-related scenarios.
3. Adding mobile-specific conditions such as backgrounding and network interruption makes the testing scope more realistic.

Scenario 07 – Dashboard / Report Generation

1. Weak Prompt

"Test the reports section."

2. What Information Is Missing?

• The prompt does not define dashboard/report context, report types, filters, date ranges or data expectations.

• It lacks empty-data behavior, export/download, permissions, failure handling, persona and output format.

3. ICEPOT Prompt

Act as a **Senior Quality Assurance Engineer** and test the **Dashboard and Report Generation functionality** of a business application.

Users should be able to view report data, apply filters, generate reports and view results based on selected criteria.

Cover correct data display, filters, date ranges, sorting, empty results, report generation, data accuracy, user access, export/download and report-generation failures. Include positive and negative scenarios. Provide **Test Case ID, Scenario, Preconditions, Test Data, Steps, Expected Result and Priority**.

4. ICEPOT Response – Summary

The response provides structured dashboard/report coverage for data accuracy, filtering, date ranges, sorting, empty results, generation, access, export and failures.

5. Three Comparison Observations

1. The weak prompt does not provide the ICEPOT information needed to define the report-testing scope.
2. The ICEPOT prompt provides a more specific response by defining data, filtering, report-generation and export-related scenarios.
3. Specifying output and audience makes the resulting test cases easier for Business Analysts, Developers and QA Engineers to understand and review.

Overall Learning

The ICEPOT framework demonstrates how additional context and structure can improve AI responses. A weak prompt gives a broad instruction and leaves important details to AI assumptions, whereas an ICEPOT prompt explicitly defines the requirements needed to produce a more specific, structured and business-aligned response.

Weak Prompt: Short instruction → Missing information → AI makes assumptions → Generic response

ICEPOT Prompt: Instruction + Context + Example + Persona + Output Format + Tone → Clear requirements → More specific and structured response

Key Takeaways

- A weak prompt may produce a useful response, but can leave important requirements to AI assumptions.
- Context makes the response more relevant to the application or business scenario.
- Persona establishes the expected professional perspective.
- Examples help the AI follow the desired structure and style.
- Output format makes the result easier to consume and review.
- Tone and audience guidance improves stakeholder understanding.
- ICEPOT transforms a generic request into a specific, structured and business-aligned prompt.